```
data_location = 'C:\Users\Hugh Williamson-Jone\OneDrive - University of Bath\Project Week 3\Project Week 3';
data_file_name = 'ESACCI-LST-L3S-LST-IRMGP_-0.05deg_1MONTHLY-20160501180000-fv1.00.nc';

% This line joins together the data location and filename to make a path for the data
data_path = fullfile(data_location, data_file_name);

% This line uses the function getnet to read in the netcdf (.nc) files into Matlab
LST = getnet(data_path);

% Extracting specific data for Land Surface Temperature (LST)
lst = LST.Data.lst(600:3000, 2600:3500) / 100;
x = LST.Data.lon(600:3000, :);
y = LST.Data.lat(2600:3500, :);

% Loop over a subset of the data to perform a simple calculation or modification
% Example: Calculate the mean LST for each row in the selected region
mean_lst = zeros(size(lst, 1), 1); % Initialize array to store mean LST for each row
for i = 1:size(lst, 1)
    mean_lst(i) = mean(lst(i, :)); % Compute the mean for each row
end

% If the average temperature is above a certain threshold, plot differently
if mean(mean_lst) > 30
    plot_title = 'High LST Area';
else
    plot_title = 'Low LST Area';
end
```

```
% Function to generate the plot
function generate_plot(x, y, lst, plot_title)
    figure()
    pcolor(x, y, lst'); shading flat;
    colormap jet
    title(plot_title)
    c = colorbar();
    c.Label.String = 'LST (°C)';
    xlabel('Longitude (°)');
    ylabel('Latitude (°)');
end

% Call the function to generate the plot
generate_plot(x, y, lst, plot_title);
```